<u>What is Process Capability Analysis?</u>

By

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I recently read the results of an American Automobile Association survey where Americans were asked to rate their own driving abilities. 73% of those surveyed considered themselves "better-than-average" drivers.[1] Obviously only 50% of the drivers can actually be better than average. So it follows that at least 23% of those surveyed are misestimating their own skills.

This over-confidence in one's abilities seems to find its way into all sorts of areas … including process capability analysis. Everyone who's been around manufacturing for any length of time has certainly heard of Cp and Cpk. Most of them know that "higher is better" when it comes to these indices. And many will nod their heads and smile when you suggest that Cpk and Ppk account for centeredness, whereas Cp and Pp do not. But only a small percentage of manufacturing professionals can cogently answer the question, "What is Process Capability analysis?"

Simply stated, process capability is a ratio between the Voice of the Customer and the Voice of the Process:

$$Process\ Capability = \frac{Voice\ of\ the\ Customer}{Voice\ of\ the\ Process}$$

The Voice of the Customer is communicated in the form of engineering prints and specifications. These documents communicate what the customer wants. For a given part feature like the flange thickness on a forged hub, the dimensional specification cited on the part's print dictates the Voice of the Customer. And part features like diameters, lengths and thicknesses specified on these engineering prints are generally assigned tolerances that limit the range of a given feature's acceptability.

The Voice of the Process communicates what range of dimensions for a given part feature the process can reliability hold. The process range is derived by measuring a random sampling of parts, then with the help of a few statistical formulas, generating the expected max and min value for the entire population of parts. The process range is the difference between those max and min values.

Rewriting our generalized formula, we can then see that,

$$Process\ Capability = \frac{Specification\ Range}{Process\ Range}$$

So in fact, higher is better when it comes to process capability. As the process range gets smaller relative to the fixed specification range, our estimate of Process Capability gets larger. A larger capability index equates to a lower probability that the process will create parts outside the specification limits. Voila! You now understand the essence of process capability.

Another layer of complexity is added to the study of process capability analysis when we consider the many process capability indices available, some of which we have already mentioned: Cp, Cpk, Pp, Ppk, Cpu, Cpl, Ppu, Ppl ... and the list continues. In my next article in this series, we will examine the key differences between these indices. But in the meantime, it is important to remember that all these indices follow the same generalized model above. In other words, they have far more in common with each other than not.

Let's consider the formula for the simplest of capability indices, Pp (pronounced "p sub p"):

$$Pp = \frac{(USL - LSL)}{6s}$$

In this formula, *USL* and *LSL* refer to the Upper and Lower Specification Limits, respectively, and *s* is a measure of dispersion called the sample standard deviation.

Decades ago, a super smart mathematician figured out that for a *normal* probability distribution of data - the bell-shaped distribution most typically encountered in manufacturing, nature, and social sciences – six times the sample standard deviation equates to about 99.7% of the values expected within the population from which that sample was drawn. And this idea that, "by a small sample we may judge of the whole piece"[2], as Miguel de Cervantes so succinctly stated, undergirds the entire field of inferential statistics of which process capability analysis is a minute part.

I live in Northeast Ohio where sugar maples trees are remarkably common. In a study I read about syrup production in the region, the heights of 112 mature sugar maple trees were measured and the resulting data analyzed. The heights of the trees measured in the study were normally distributed with an arithmetic mean of 64.5 feet and a standard deviation of 6.25 feet. Using this data and what we now know about the normal distribution, we can infer that the expected range of heights among of mature sugar maple trees in Northeast Ohio is 37.5 feet (6 * 6.25 feet).

Since we know the mean tree height is 64.5 feet tall, we can equally distribute this range above and below it to predict the heights between which 99.7% of the mature sugar maple trees in this region will measure:

$$64.5 + (3 \times 6.25) = 83.25 \text{ feet}$$

$$64.5 - (3 \times 6.25) = 45.75 \text{ feet}$$

This is quite similar to the procedure used in manufacturing to conduct a process capability study:

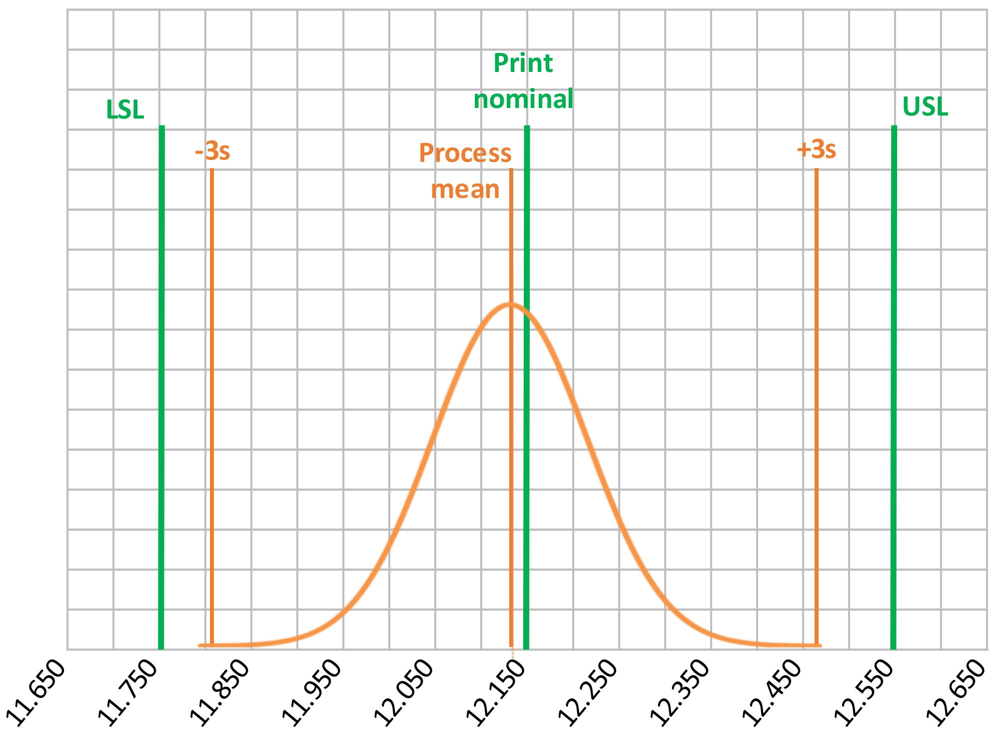
1. Draw a sample of parts from the population of parts under study.
2. Measure the feature in question on each sample.
3. Verify the data is normally distributed.
4. Calculate the average and standard deviation.
5. Calculate predictions about max and min values in the population from which the sample came.
6. Compare those predictions to the specification limits.

The obvious difference between manufactured components and maple trees is the specification limits. Trees have no specs, but of course manufactured parts do.

Imagine we used the above procedure to examine the thickness of a forged flange. We could start by randomly selecting a sample of forgings, say 100 pieces, from a much larger production run. We could then measure the flange in question of all 100 pieces and record our flange thickness values in Excel.

By applying Excel's *average* and *stdev.s* functions to our array of thickness data points, we could readily find the sample data's arithmetic average (also called *mean*) and sample standard deviation, the statistics needed to determine the Voice of the Process.

Imagine that from our dimensional study, we calculate a mean of 12.140 mm and a standard deviation of .110 mm. Assuming our measured values were normally distributed, we can then estimate the process range at .660 mm (6 x .110mm). If the print specification is 12.150 +/- .400 mm, our specification range is .800 mm. Plotting this together, we find the following:



Using our capability formulas,

$$Process\ Capability = \frac{Specification\ Range}{Process\ Range}$$

Pp = .800 mm / .660 mm = 1.21

Notice that Pp takes no account for the centeredness of the process within the specification limits. In this example, our flange thickness may be the length of a tennis court with all the parts out of specification, but with a standard deviation of .110 mm and a specification range of .800 mm, our Pp will be 1.21.

So why bother with Pp at all? Because with it we can still draw some important conclusions about our process. The most important conclusion we can draw in our example is that the specification range is 1.21 times bigger than our process range. Therefore, *if* the process were centered within the specification range, a very high percentage of the total parts in the population would remain within the specification limits. That's an important detail to consider.

Referring back to formula,

$$Pp = \frac{(USL - LSL)}{6s}$$

we can see that a Pp of 1.0 requires identically sized process and specification ranges, leaving no room for off-centeredness where parts begin to transgress either the upper or lower specification boundaries. From this we can conclude that a higher Pp "buffers" a certain amount of off-centeredness, lowering the probability of out-of-specification parts.

It is at this concern over the centeredness of a process within its specification limits where Ppk can serve us well. Let's look at the formula:

$$Ppk = \min\left[\frac{USL - \bar{x}}{3s}, \frac{\bar{x} - LSL}{3s}\right]$$

The first thing we notice is that Ppk is the minimum value of two different equations. So when calculating these indices, only half your effort is utilized since one of the equations is not used in the final value.

Incidentally, here we also encounter two additional capability indices, Ppu and Ppl, as in "Pp Upper" and "Pp Lower", the first and second halves of the above equation, respectively.

The second thing you'll notice is that the two equation halves have the same denominator, $3s$. This we already know as half the process range. So the equation with the smaller numerator generates the capability index while the equation with the larger numerator falls away.

Considering then just the numerators, (USL – mean) and (mean – LSL), the smaller of the two is always the one with the shorter distance between the process mean and the nearest specification limit.

Qualitatively speaking, given that the process range is equally distributed on both sides of the process mean, if the process mean is closer to the upper specification limit than the lower, then it's more probably that parts will be out of specification on the upper side instead of the lower side of the range. Naturally, the opposite is also true if the process mean is closer to the lower specification limit. This idea that parts are more likely to transgress one specification limit rather than the other is what justifies Ppk's sole focus on the smaller of the two numerators.

Therefore, where Pp is the ratio of the entire specification range for a given characteristic to its entire process range, Ppk is a ratio of half the process range to the distance between the process mean and closest specification limit.

Let return to our forging example:

Process average = 12.140 mm

Sample standard deviation = .110 mm

Print specification = 12.150 +/- .400 mm

USL = 12.550 mm

LSL = 11.750 mm

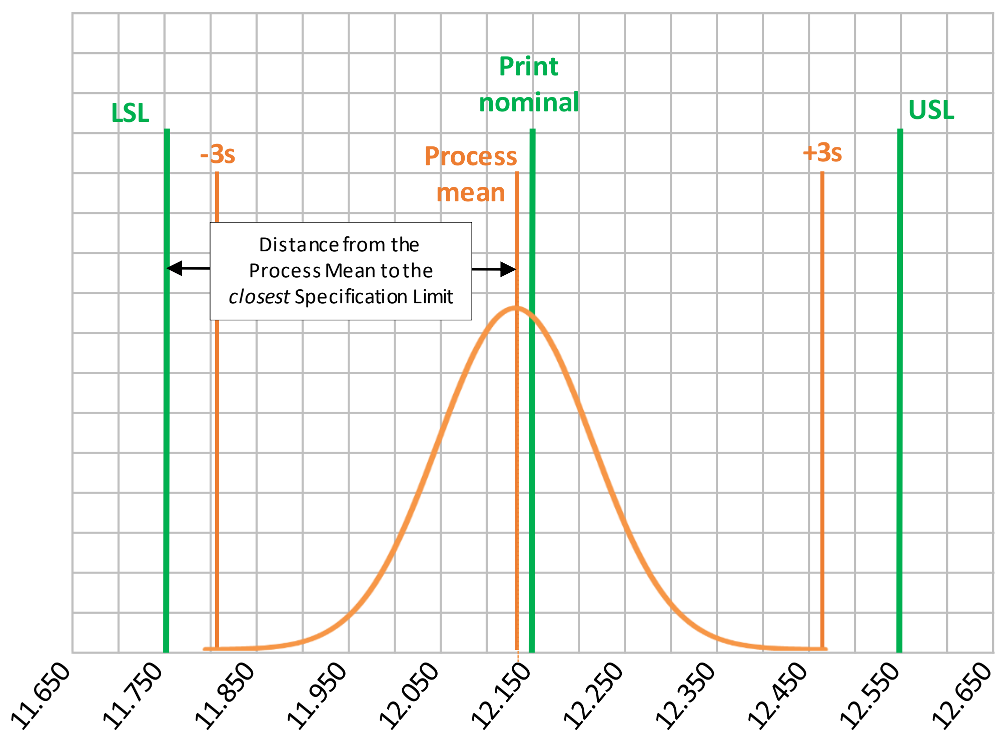
Feeding these numbers into our Ppk equation, we obtain:

$$Ppk = \min\left[\frac{USL - \bar{x}}{3s}, \frac{\bar{x} - LSL}{3s}\right]$$

$$Ppk = \min\left[\frac{12.550 - 12.140}{3(.110)}, \frac{12.140 - 11.750}{3(.110)}\right]$$

$$Ppk = \min[1.24, 1.18] = 1.18$$

From this calculation of Ppk, we can infer that the distance from the process mean to the closest specification limit is 1.18 times larger than half the process range. Logically then, we can conclude that as Ppk get larger, the probability of the far-reaching ends of the process range transgressing a specification limit gets smaller.



Here are a few other logical conclusions you can draw from these capability formulas:

- If a process is perfectly centered within specification limits, Pp = Ppk.
- In any given study, Ppk can never be larger than Pp.
- No one capability index is better than the others. They work together to describe a process.

And for those wondering about the more familiar capability indices, Cp and Cpk, in the next article in this series, I will explain the differences between those indices and the ones we discussed in this article.

But for now, if someone asks you, "How good is your understanding of process capability analysis?", you can confidently respond, "better than average."

References:

1. Koppel, L., Andersson, D., Tinghög, G., Västfjäll, D., & Feldman, G. (2023). We are all less risky and more skillful than our fellow drivers: Successful replication and extension of Svenson (1981). *Meta-Psychology*.
2. de Cervantes, M. (1992). Don Quixote (P. A. Motteaux, Trans.). Wordsworth Editions.

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